

The Sound of a Sonnet: Poem-to-Song Recommendation with Pseudo-Supervision

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Overview

We aim to predict the most musically compatible songs for a given poem by modeling cross-domain semantic and affective similarity between lyrics and poetic text. Because no labeled poem–song datasets exist, we develop a retrieval model trained using pseudo-supervised labels derived from similarity heuristics. Our system embeds poems and song lyrics into a shared latent space via a dual-encoder architecture optimized with contrastive learning. Using human triplet evaluation, the model retrieves songs that align with poetic tone, structure, and affect. Our best variants approach human-level agreement while outperforming MPNet-only baselines.

Data

Our dataset contains **3,413 poems** from PoetryDB and **2,934 song lyrics** collected from Genius and Spotify metadata^[5–7]. We clean and normalize all text by removing formatting artifacts (e.g., stanza labels, duplicated sections), lowercasing, removing markup, and standardizing Unicode.

Poem (cleaned)	Song (cleaned)
• title: "A Song of Autumn" • author: "Adam Lindsay Gordon" • lines_clean: ["where shall we go for our garlands glad", "at the falling of the year", ...] • linecount: 16	• title: "Father Figure" • artist: "Taylor Swift" • album: "The Life of a Showgirl" • duration_ms: 212777 • lyrics_clean: ["when i found you you were young wayward lost in the cold", ...]

Figure 1. *

Cleaned poem vs. song lyric inputs

Features

Each poem/song is represented using **810 total dimensions**, combining (1) MPNet sentence embeddings, (2) semantic attributes from zero-shot emotion/theme classifiers (36 dims), and (3) lexical + structural metrics (6 dims). All features are standardized and concatenated prior to encoder projection.

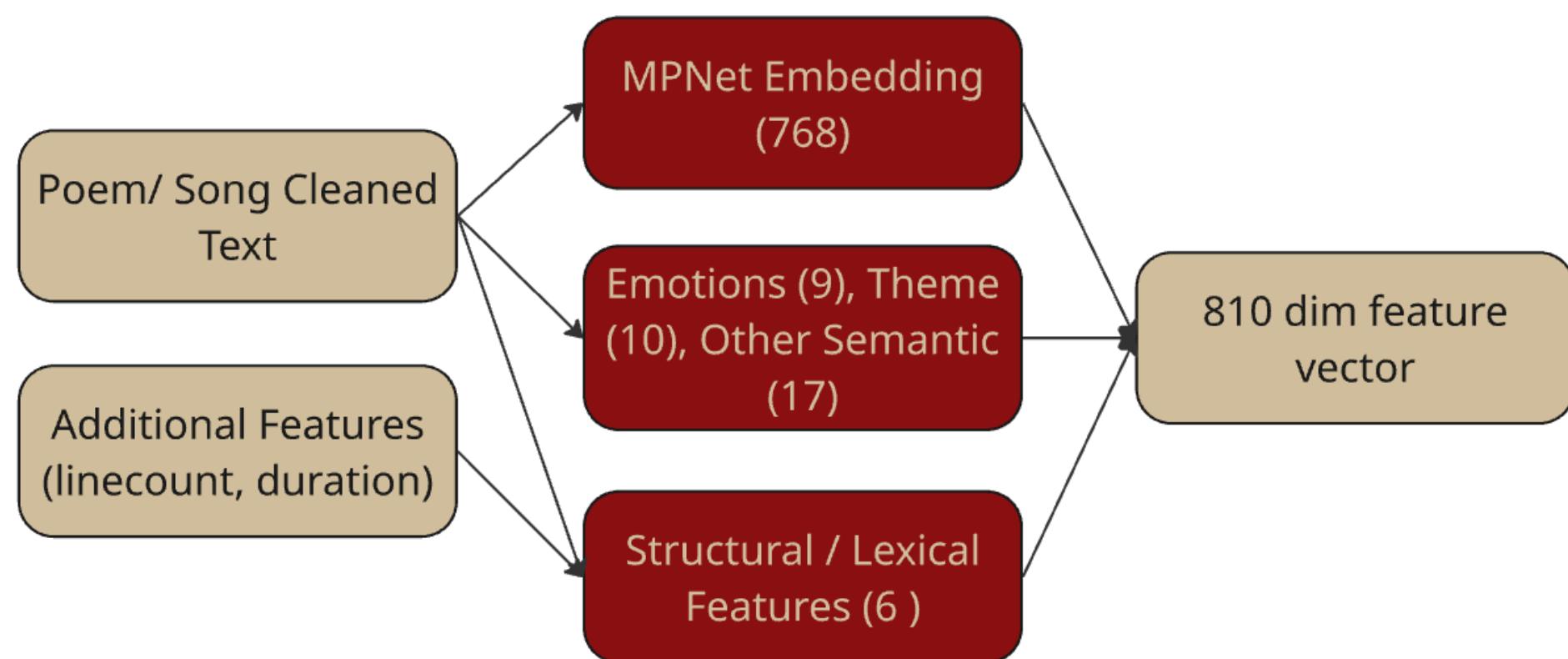


Figure 2. *

810-D representation combining MPNet, semantic attributes, and structural/lexical features

Models

We train a **shared dual-encoder** that projects poems and songs into a **128-dimensional embedding space** using feature-specialized MLP branches followed by a projection and L2 normalization. Supervision is obtained automatically by estimating poem–song relatedness with a **weighted cosine similarity** over MPNet, affective/theme features, and structural/lexical attributes:

$$\text{Sim}(p, s) = \alpha \cos(\mathbf{x}^{mp}) + \beta_{emo} \cos(\mathbf{x}^{emo}) + \beta_{theme} \cos(\mathbf{x}^{theme}) + \beta_{other} \cos(\mathbf{x}^{other}) + \gamma \cos(\mathbf{x}^{feat}).$$

Top- k pairs under this heuristic serve as pseudo-positives, while the lowest 5% are treated as pseudo-negatives. These pseudo-labels train the encoder using **symmetric InfoNCE contrastive loss**^[3], and retrieval is performed by **cosine similarity** in the learned space. Models are trained using PyTorch and HuggingFace Transformers^[8,9]

Dual Encoder Architecture

Five distinct branches capture complementary linguistic signals in our dual encoder: deep semantic content (MPNet), affective tone (emotion), thematic constructs, additional stylistic semantics, and structural/lexical statistics. Branch outputs are concatenated, normalized, linearly projected to 128-dimensional embeddings, and L2-normalized.

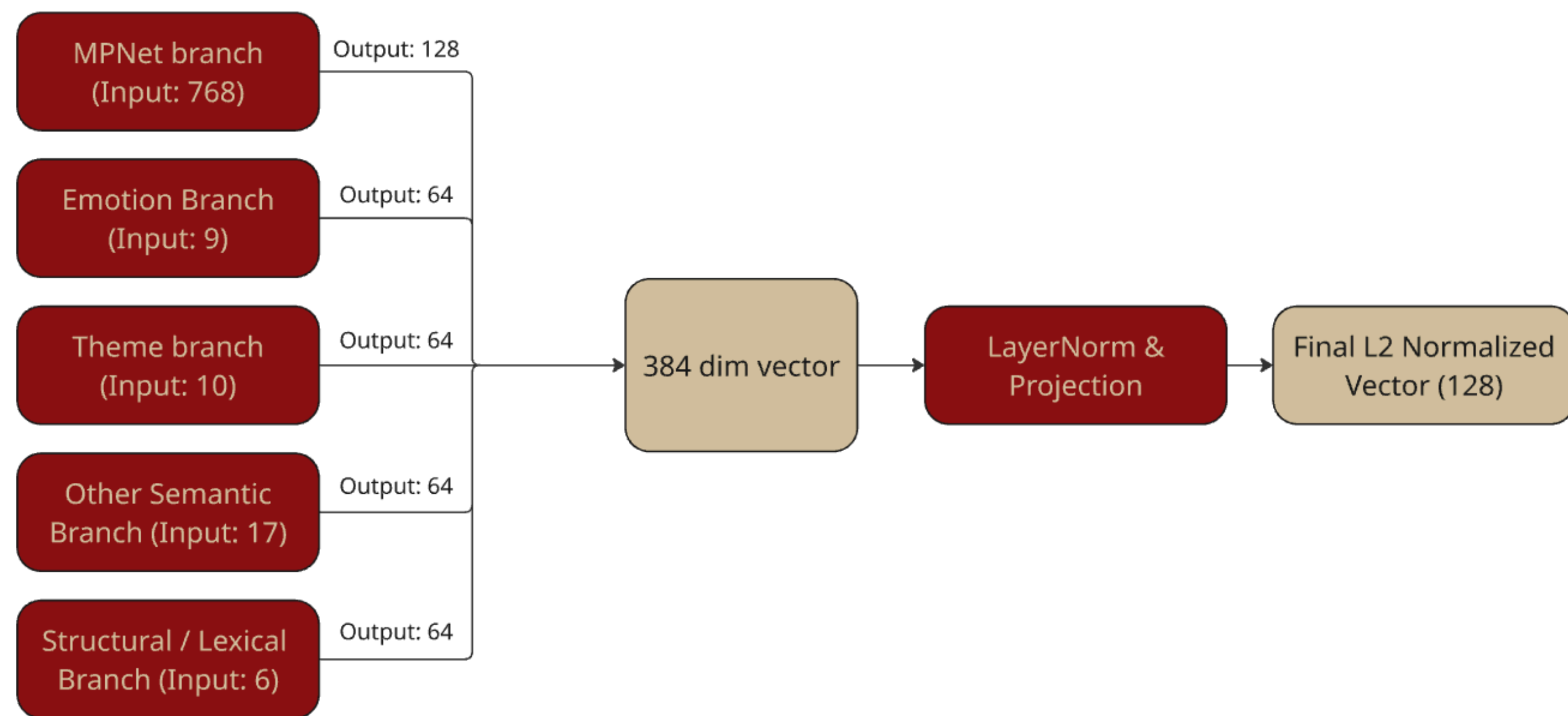


Figure 3. *

5 branch dual encoder architecture to 128-D embedding

Contrastive Learning with InfoNCE

We use a symmetric contrastive objective to encourage poem–song pairs to have high cosine similarity, while unrelated pairs are pushed apart. Given a pseudo-positive pair (p_i, s_i) and a batch of size B , we compute a scaled similarity matrix:

$$\ell_{ij} = \frac{\mathbf{z}_{p_i}^T \mathbf{z}_{s_j}}{\tau},$$

where τ is a learnable temperature parameter. The loss for ranking songs given a poem is:

$$\mathcal{L}_{p \rightarrow s} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\ell_{ii})}{\sum_{j=1}^B \exp(\ell_{ij})},$$

and the loss for ranking poems given a song is defined analogously. The final InfoNCE objective averages both directions:

$$\mathcal{L} = \frac{1}{2} (\mathcal{L}_{p \rightarrow s} + \mathcal{L}_{s \rightarrow p}).$$

Evaluation and Results

We evaluate using a human triplet ranking task: annotators are shown a poem and two candidate songs and choose the better match based on tone, theme, and style. A model prediction is correct when its cosine ranking agrees with the human choice. To estimate an achievable ceiling, a second annotator labels the same triplets; their agreement ($\sim 80\%$) defines the level at which even humans disagree and thus bounds model performance.

Our best feature-aware encoders reach ≈ 78 -80% agreement, matching human consistency and outperforming a strong MPNet cosine baseline ($\approx 67\%$). Models emphasizing semantic and structural cues perform best, suggesting poetic–musical similarity depends on affect and stylistic structure over literal text matching. Even without picking the best model, average accuracies surpass MPNet and near human levels, providing a conservative lower bound that still beats MPNet.

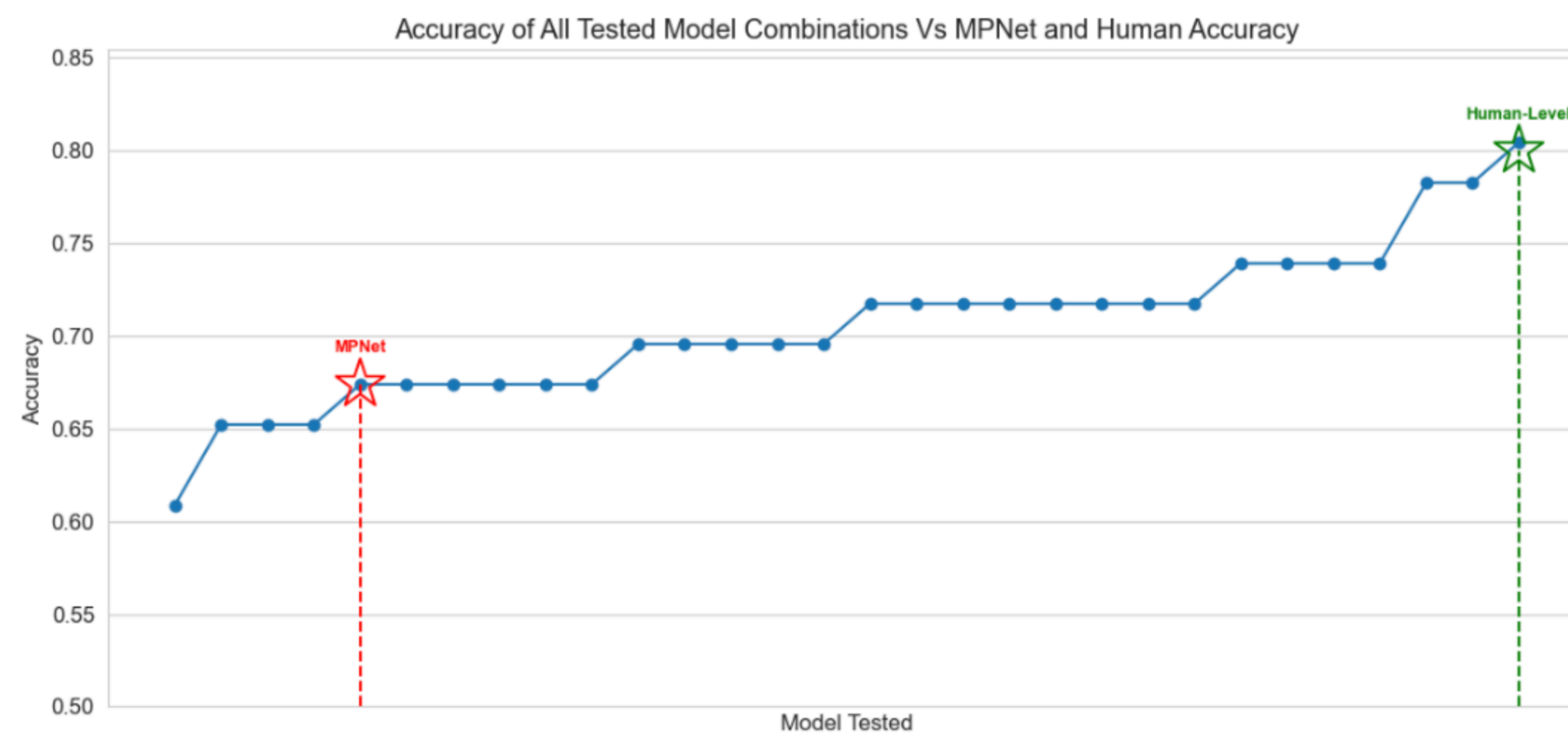


Figure 4. *

Accuracy of all tested poem–song retrieval models

Discussion and Future Work

Feature-aware models show that poem–song similarity depends more on emotional tone and stylistic structure than literal word overlap. Most errors stem from ambiguous or mixed poetic intent, suggesting subjective difficulty rather than model failure. Pseudo-supervision is effective, but heuristic labels limit the upper bound of performance.

Future Work

Future work includes collecting a larger human-labeled dataset to reduce noisy evaluation and enable semi-supervised refinement of the embedding space. We also plan to integrate **audio features** (rhythm, timbre) to capture musical alignment beyond text.

References

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